

→ Normal distribution → 68-95-99.7 rule

$$X \sim N(\mu, \sigma)$$

Given ~~x~~ → $x_1, x_2, x_3, \dots, x_n$

income

Q. If X Gaussian distribution or ND

income

$D_x \rightarrow$ income →



~~x~~



KS
AD
 ~~$\sigma = \sigma$~~

ND

OR NOT

Q-Q plot
quantile
quantile
statistical
testing
(KS, AD)

C_1	10K	7K
C_L	9K	9K
C_S	7K	12K

Q-Q

$X \sim x_1 x_2 x_3 \dots x_{500}$



Sort x_i (increasing order) and compute percentile

input

→ $x_1 x_2 x_3 \dots x_{500}$

sorting

→ $x'_1 x'_2 x'_3 \dots x'_{500}$

compute percentile

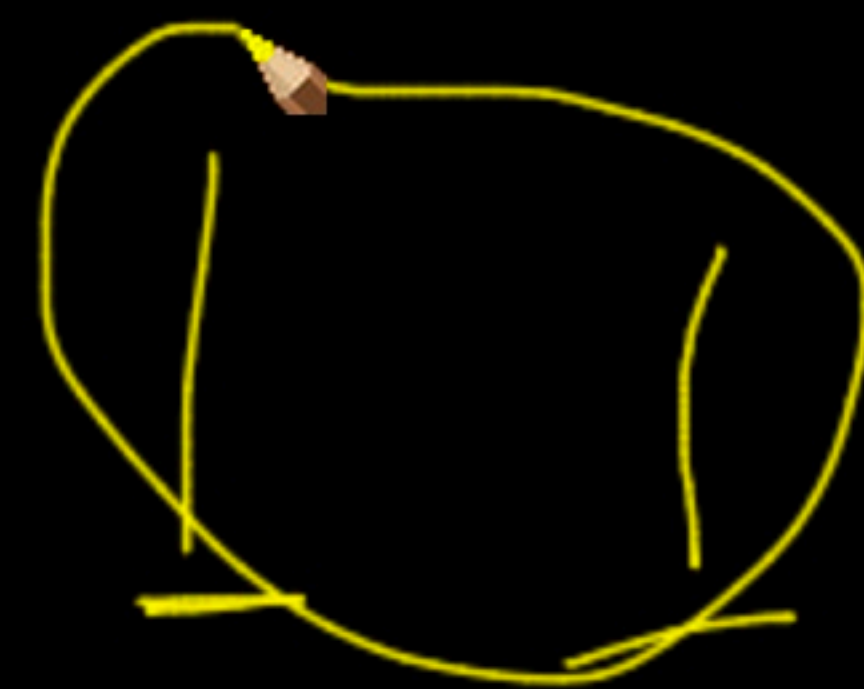
x'_5 x'_{10} x'_{15} x'_{20}

100th

x'_{500}

$x'_i \rightarrow i^{th}$ percentile value of x

~~4~~ $x^{(1)}$ $x^{(2)}$ $x^{(3)}$ — — — $x^{(100)}$



(2) $y \sim N(0, 1) \rightarrow$ Std Normal distribution

$y_1, y_2, y_3, \dots, y_{1000}$

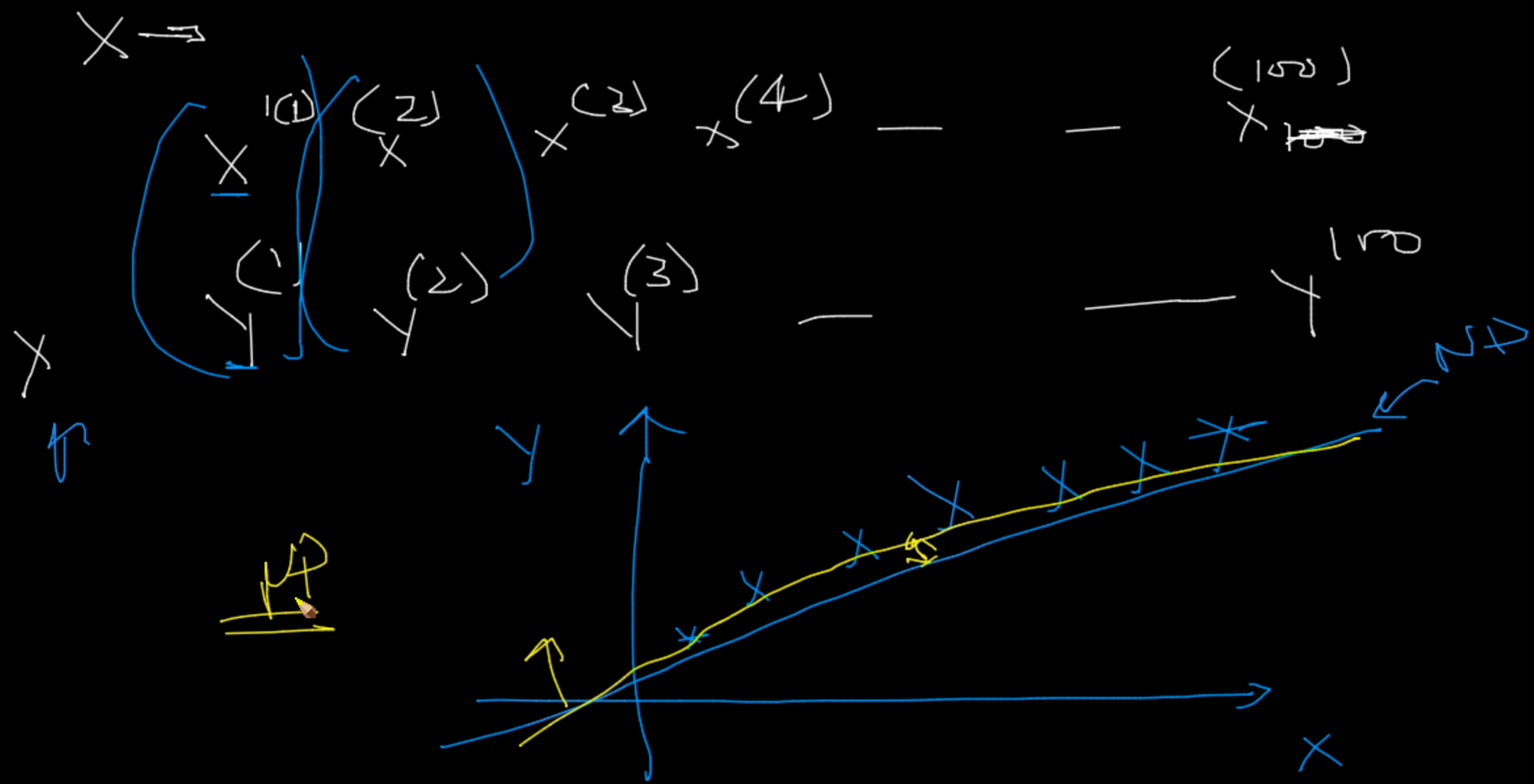
sort it

$y_1', y_2', y_3', \dots, y_{1000}'$

percentile compute

$y_{1000}' \rightarrow y^{(1)}, y^{(2)}, \dots, y^{(100)}$

Step 3



A new page has been created



Quantifying Relationship among feature

2 feature $\rightarrow$ height and weight

Q $\rightarrow$ Are they co-related? i.e. $h \uparrow \rightarrow w \uparrow$ or $w \downarrow$

ex Retail

discount

sales

discount & sales

10

200

15

300

1

1

1

$d \uparrow$ ~~$w \uparrow$~~ ~~$w \downarrow$~~

$\boxed{\text{sales} \uparrow \downarrow}$

domain $\rightarrow$ $\frac{1}{n}$

$$\begin{array}{l} \uparrow \underline{F_1 \propto F_2} \uparrow \\ F_1 \propto \frac{1}{f} \end{array}$$

$\Rightarrow$

37 MEASURE

SIMPLE

- 1) ~~CO~~ - variance $\nearrow$
- 2) Pearson's co-relation coeffⁿ
- 3) Spearman's

co-variance

$\bar{x}$ $\bar{y}$

→ $\text{covariance}(x, y)$

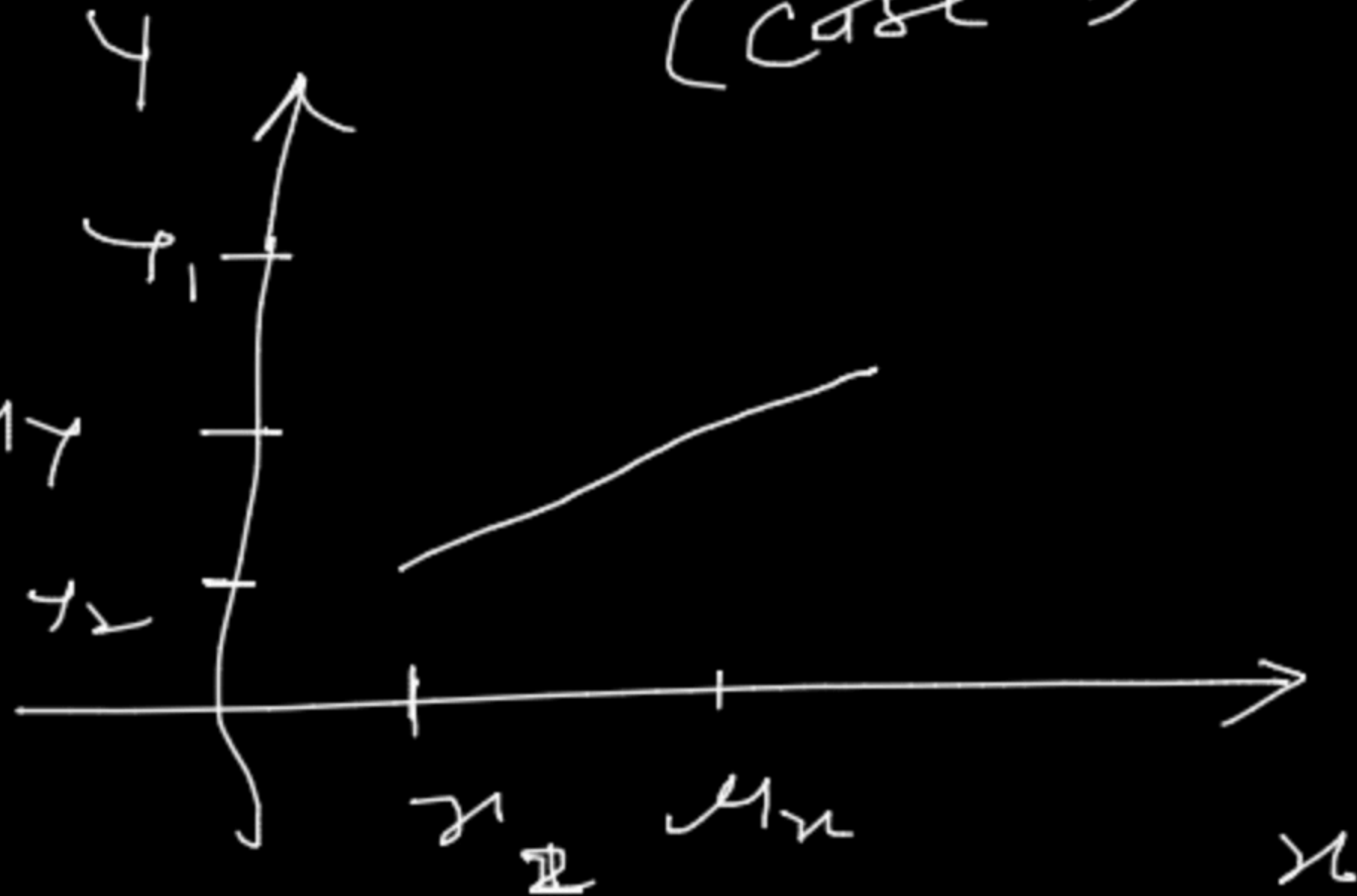
$$= \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x) * (y_i - \mu_y)$$

covariance($\overset{\uparrow}{x}, \overset{\uparrow}{x}$) = variance(x)

$$\frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)(x_i - \mu_x)$$

(x, y_1)

(case -i)



$$\frac{(x - \mu_x)}{\sigma_x} \times \frac{(y_1 - \mu_y)}{\sigma_y}$$

$(+)$ $(+)$

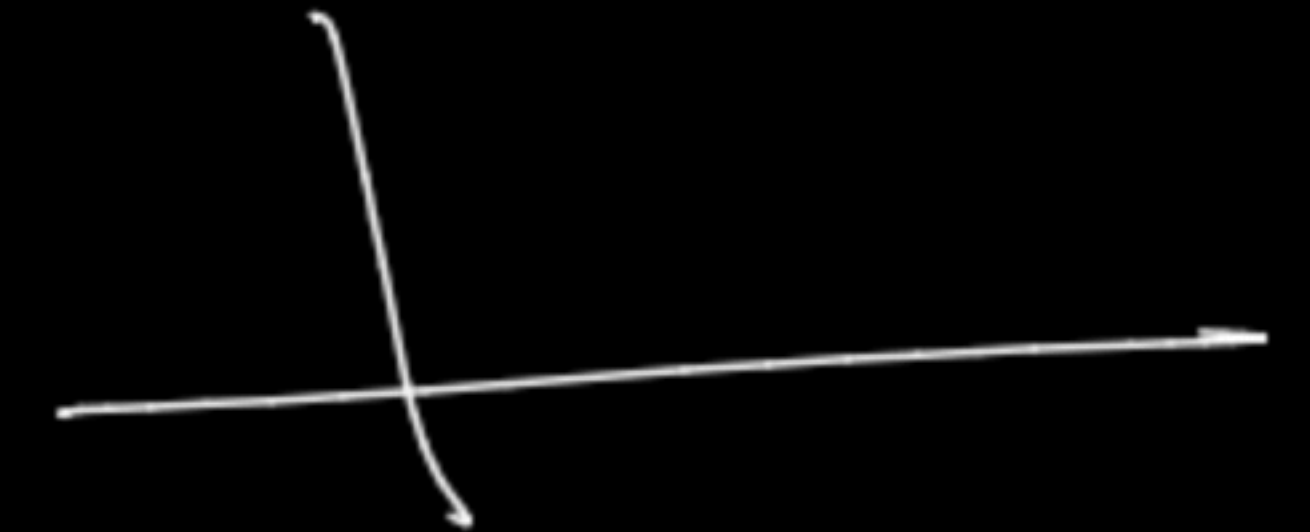
$x \uparrow \quad y \uparrow$

0.2
 $y \rightarrow ?$

$(-)$

$(-)$ $\rightarrow$

$$\text{COV}(x, y) \rightarrow (+ve)$$



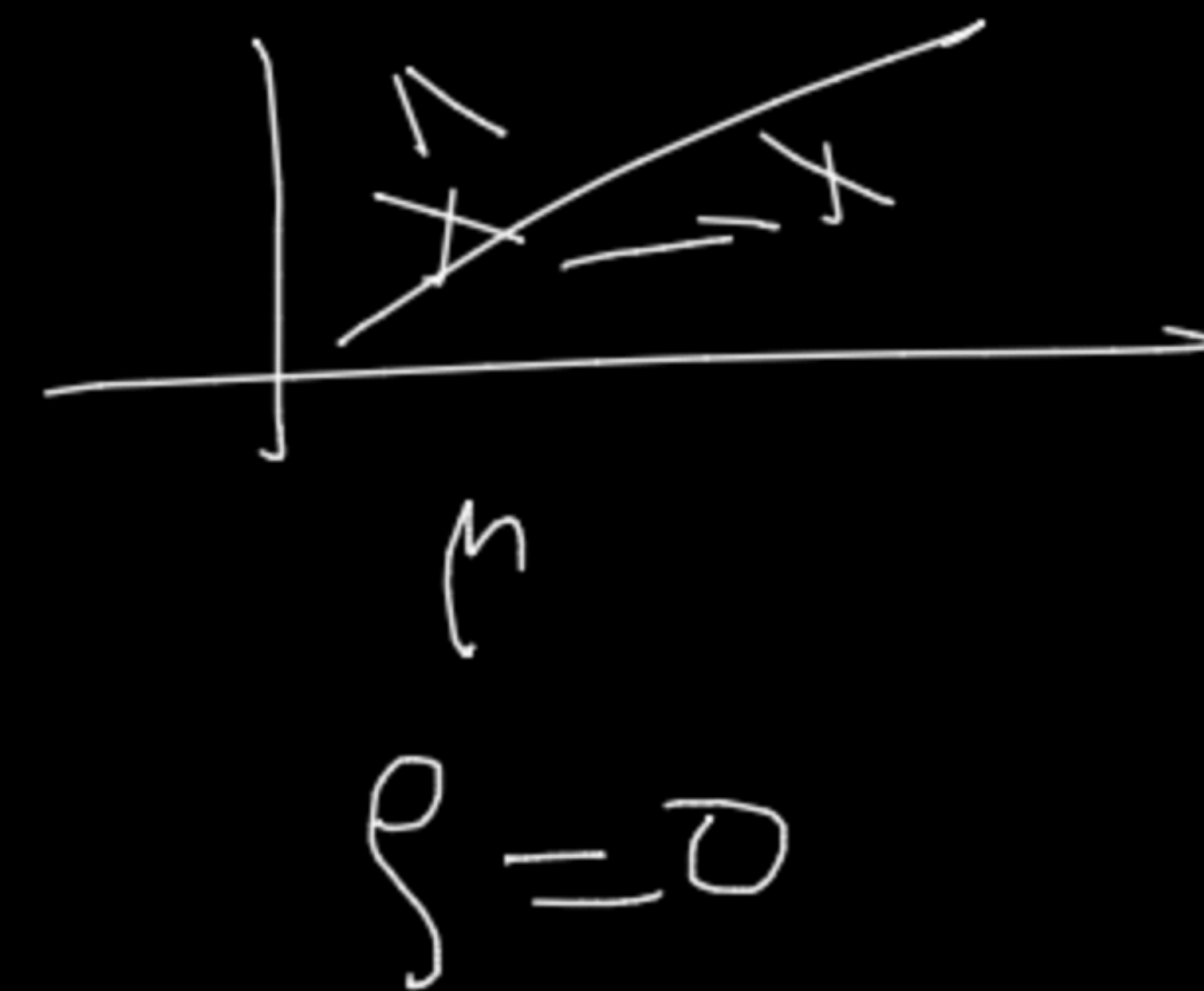
(case ii)

$$(+)$$
 $\times$ $(-)$ $\Rightarrow$ $-ve$

$$(-)$$
 $\times$ $(+)$ $\Rightarrow$ $-ve$

$$\text{COV}(y, y) = (-ve)$$

$$\rho_{x,y} = \frac{\text{cov}(x,y)}{\sigma_x \sigma_y} \leftarrow \text{formula}$$

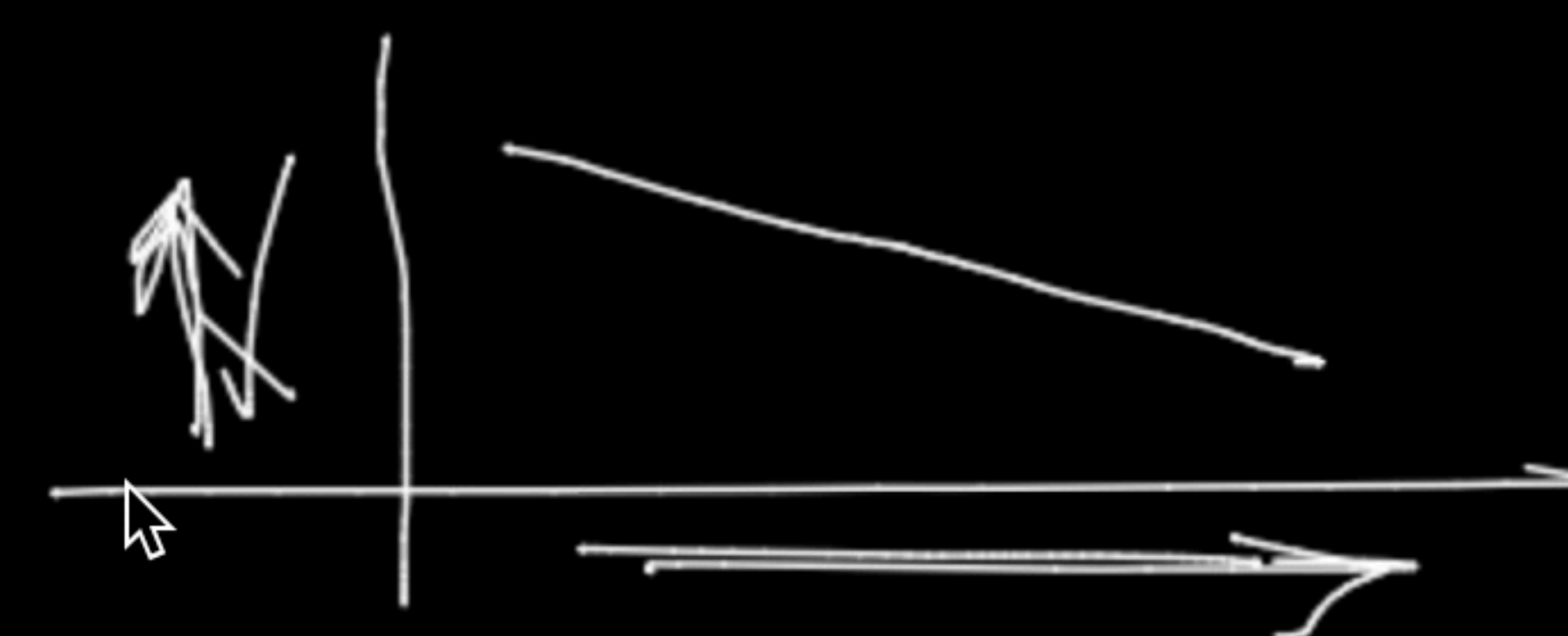
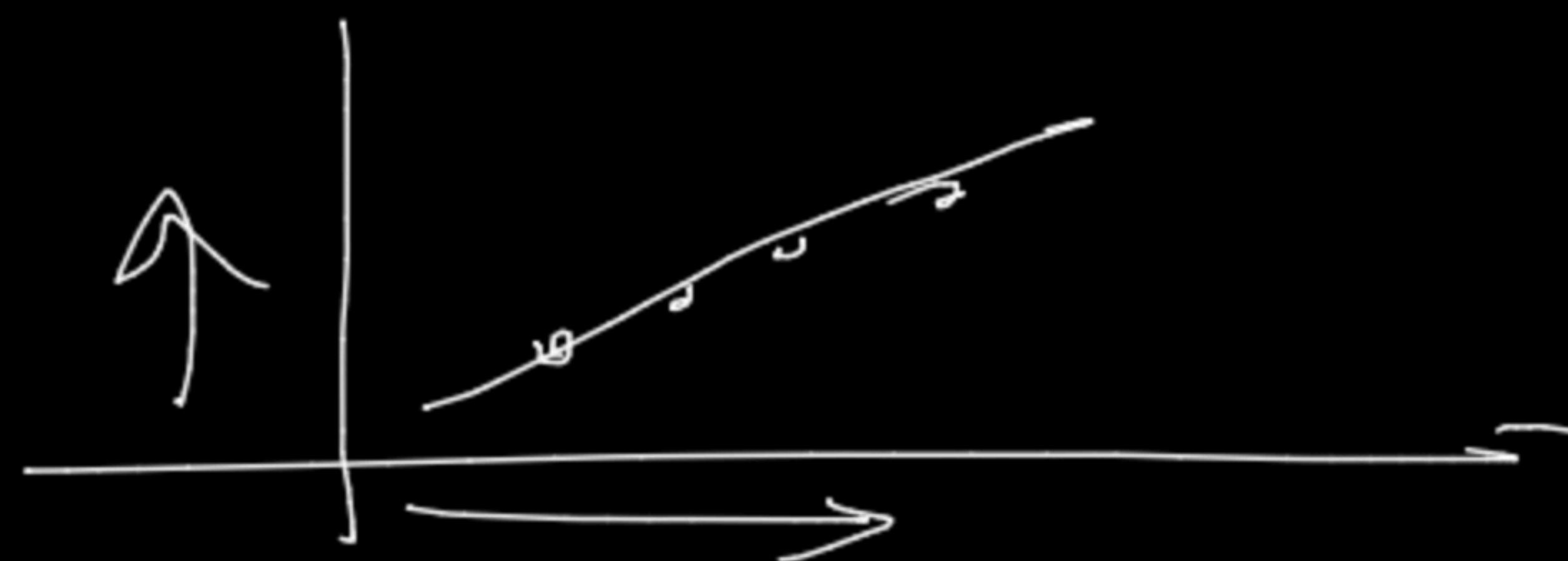


$$-1 \leq \rho < 1$$

$$\frac{\text{cov}(x,y)}{\sigma_x \sigma_y}$$

$$\rho = 1$$

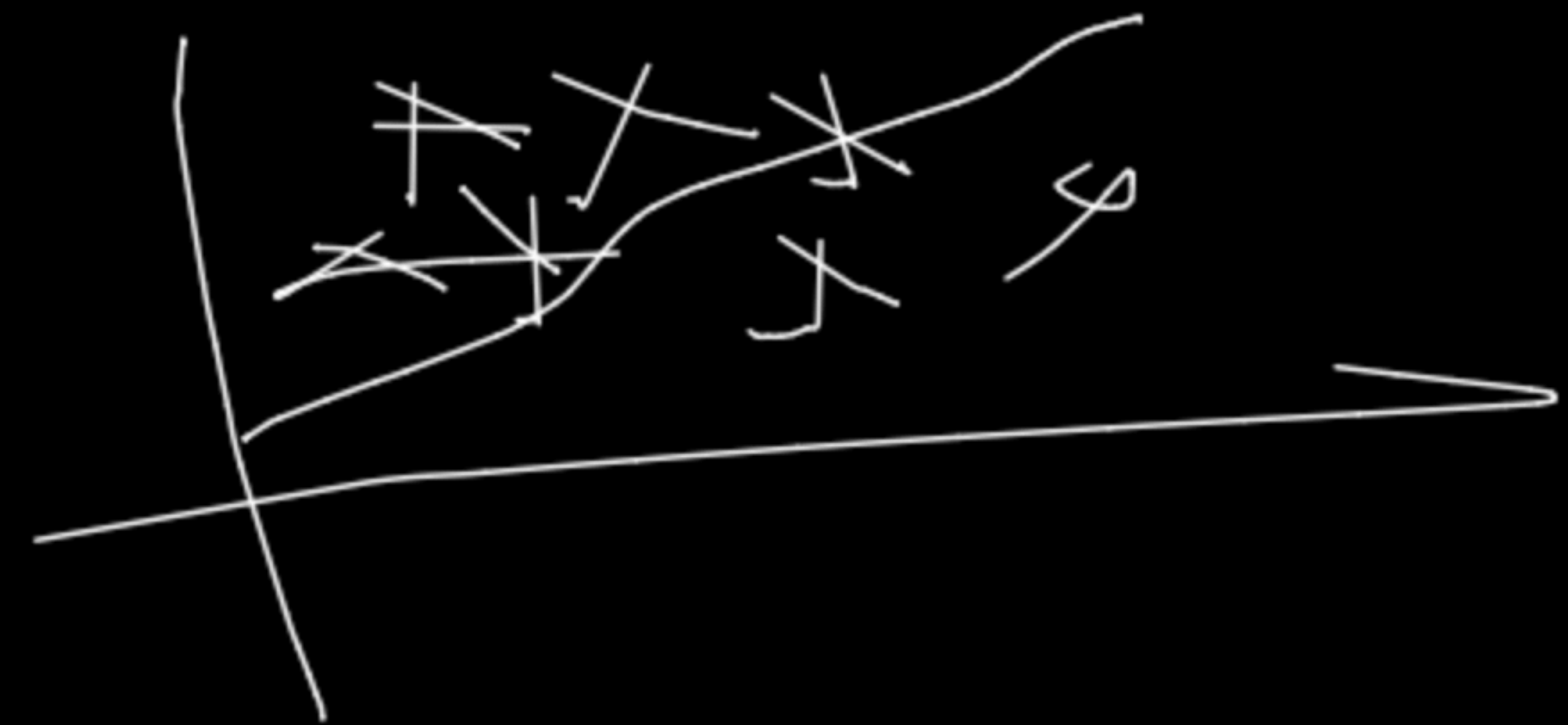
$$\rho = -1$$



limitations

only +ve when linear relation b/w
x and y

16 $y = x^2$



x $\rightarrow$

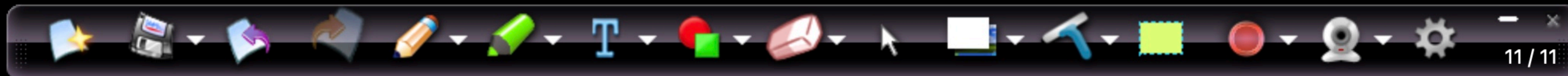


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